Segmentation on Medical mages

Total points 9

1. Which of the following is a segmentation task? **1 point**

- Determining whether there is a mass in a chest X-ray
- Determining whether a brain tumor is present in an MRI
- Determining which areas of the brain have tumor from an MRI
- None of the above

2. What is the MAIN disadvantage of processing each MRI slice independently using a 2D segmentation model (as mentioned in the lecture)? **1 point**

Hint: watch the lecture video "Segmentation" to help you answer this question.

- It is difficult to register slices of MRI models
- You lose some context between slices
- None of the above
- 3D models are always better than 2D models

3. The U-net consists of... **1 point**

- Just a contracting path
- Just an expanding path
- An expanding path followed by a contracting path
- A contracting path followed by an expanding path

4. Which of the following data augmentation is most effective for MRI sequences? **1 point**

- ◯ Randomly shuffle the pixels in each slice
- ◉ Rotation
- ◯ Shifting each pixel to the right by a constant amount with wrap around
- ◯ Shuffling the slices

5. What is the soft dice loss for the example below? **1 point**

$$L(P,G) = 1 - \frac{2\sum_{i=1}^{n} p_i g_i}{\sum_{i=1}^{n} p_i^2 + \sum_{i=1}^{n} g_i^2}$$

P

0.3	0.7	0.3
0.7	0.9	0.7
0.3	0.7	0.3

G

0	1	0
1	1	1
0	1	0

i	p_i	g_i	$p_i g_i$	p_i^2	g_i^2
1	0.3	0	0	0.09	0
2	0.7	1	0.7	0.49	1
3	0.3	0	0	0.09	0
4	0.7	1	0.7	0.49	1
5	0.9	1	0.9	0.81	1
6	0.7	1	0.7	0.49	1
7	0.3	0	0	0.09	0
8	0.7	1	0.7	0.49	1
9	0.3	0	0	0.09	0

$\sum p_i g_i = 3.7$

$\sum p_i^2 = 3.13$

$\sum g_i^2 = 5$

$L(P,G) = \frac{2 \times 3.7}{3.13 + 5} = \frac{7.4}{8.13} \rightarrow \boxed{0.91}$

$1 - 0.91 = 0.089$

- ◯ 0.910
- ◉ 0.089
- ◯ -0.089
- ◯ 0.544

6. Look at the output of model 1 and model 2: **1 point**

P1

0.3	0.7	0.3
0.7	0.9	0.7
0.3	0.7	0.3

P2

0.5	0.7	0.5
0.7	0.9	0.7
0.5	0.7	0.5

G

0	1	0
1	1	1
0	1	0

Which one will have a lower soft dice loss?

$1 - \frac{2 \times 0.37}{2.77 + 5} = \frac{0.74}{7.77} = \boxed{0.9}$

Hint: Notice the prediction scores of P1 and P2 on the pixels where the ground truth is 1. This may help you focus on certain parts of the soft dice loss formula:

$$L(P, G) = 1 - \frac{2\sum_{i=1}^{n} p_i g_i}{\sum_{i=1}^{n} p_i^2 + \sum_{i=1}^{n} g_i^2}$$

P_2 $\sum p_i g_i = 0.37$ $\sum p_i^2 = 2.77$ $\sum g_i^2 = 5$

i	p_i	g_i	$p_i g_i$	p_i^2	
1	0.5	0	0	0.25 ✓	1.0
2	0.7	1	0.7	0.49 ✓	1.96
3	0.5	0	0	0.25 ✓	+
4	0.7	1	0.7	0.49 ✓	0.81
5	0.9	1	0.9	0.81	1.97
6	0.7	1	0.7	0.49 ✓	0.8
7	0.5	0	0	0.25 ✓	2.77
8	0.7	1	0.7	0.49 ✓	
9	0.5	0	0	0.25 ✓	

- (selected) Model 1 has a lower loss
- None of the above
- Model 2 has a smaller loss
- They will be the same

7. What is the minimum value of the soft dice loss? **1 point**

$$L(P, G) = 1 - \frac{2\sum_{i=1}^{n} p_i g_i}{\sum_{i=1}^{n} p_i^2 + \sum_{i=1}^{n} g_i^2}$$

$\lim_{p \to 0^+} \frac{+\infty}{+\infty +} \approx 1$

$1 - \approx 1 = 0$

$1 - \lim_{p \to -\infty} \frac{-\infty}{-\infty} \approx +\infty$

- (selected) 0
- - infinity
- 1
- 4

8. An X-ray classification model is developed on data from US hospitals and is later tested on an external dataset from Latin America. Which if the following do you expect? **1 point**

- Performance remains unchanged
- Performance improves on the new dataset
- None of the above
- Performance drops on the new dataset

9. Which of the following is an example of a prospective study? **1 point**

- A model is trained and tested on a dataset of X-rays collected between 2001 and 2010
- None of the above
- A model is trained on data collected between 2001 and 2010 and then validated on data collected between 2011 and 2013
- A model is deployed for 1 year in an emergency room and its performance over that time is evaluated

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